

WISHBONE INTERFACE

WISHBONE system-on-chip interconnect defines two types of interfaces, called MASTER and SLAVE interfaces. MASTER interfaces are cores that are capable of generating Bus cycles. SLAVE interfaces are cores that are capable of receiving Bus cycles.

WISHBONE MASTER/SLAVE MODULE OPERATION

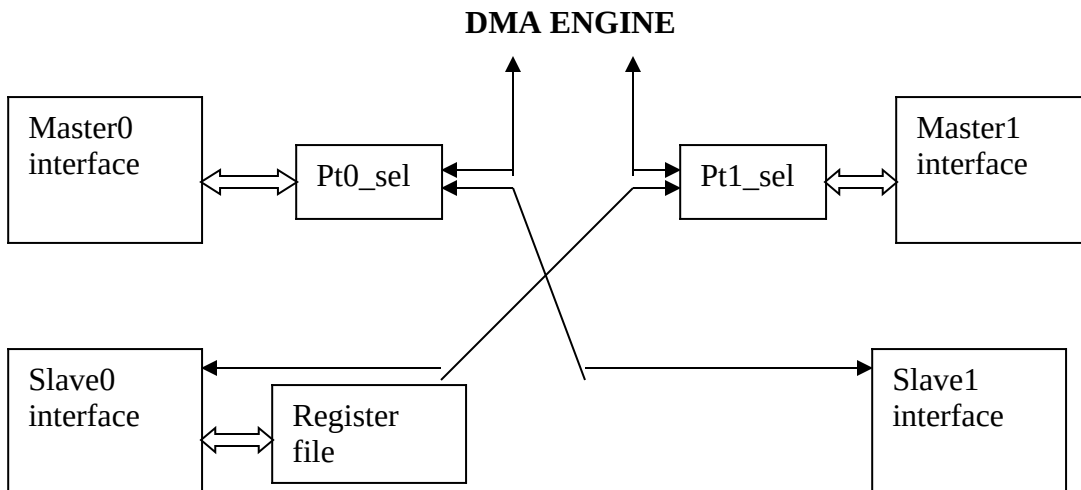
Wishbone MASTER: Wishbone Master module role is to make appropriate bus cycles depending on the signals from the core. MASTER interfaces are cores that are capable of generating Bus cycles.

Wishbone SLAVE: Wishbone slave module role is to transfer the data to/from the memory/peripheral and report the result of the transfer to the master module. Slave interfaces are cores that are capable of receiving Bus cycles.

Wishbone Master and Slave transfer data in both the DMA operation and PASS through operation. In DMA operation register file is selected.

Pass through operation: In pass through operation this core acts as a Bridge. It does not add any functionality. Pass-through logic is combinatorial only (i.e. in pass through mode signals are not latched).

When instantiating the core user passes the register file address (rf_addr) of 4-bit value. This 4-bit value is compared to the WISHBONE address[31:28]. If it matches, then the internal register file of the DMA is selected otherwise the DMA will operate in Pass through mode.



In Pass through mode signals coming out from master and Slave are:

Master and Slave Pass through select signal: Slave pass through signal is asserted HIGH when rf-addr of 4-bit parameter value does not match the Wishbone address of [31:28] bits. When select signal is HIGH indicates Pass through mode is selected.

Master Pass through data input: Master Pass through data input is a combination of wishbone master output signals i.e. data output, address, sel, we(write enable), cyc(cycle) and strobe.

Master Pass through data output: This is a combination of wishbone master input signals i.e. master data input, acknowledge, error, retry.

Slave Pass through data input: Slave Pass through data input is a combination of wishbone slave input signals i.e. data input, address, sel, we(write enable), cyc(cycle) and strobe.

Slave Pass through data output: This is a combination of wishbone slave output signals i.e. slave data output, acknowledge, error, retry. Pass through output data of slave (slv_pt_out[70:0]) is given to master mast_pt_in([70:0]). Pass through output of master(mast_pt_out[34:0]) is given to slave input slv_pt_din[34:0]).

Wishbone in DMA operation

In this DMA operation Register file is selected and pass through signals will not come in picture. Individual master and slave signals are used for data transfer.

Wishbone READ/WRITE CYCLE

DMA interface protocol with WISHBONE Master interface

To start the data transfer DMA engine gives start signal dma_wbif_mast0_start or dma_wbif_mast1_start wishbone Master interface0 or wishbone master interface1 depending on channel csr[2] and csr[1] bit to initiate the cycle. DMA also asserts dma_wbif_mast0_wrnrdr or dma_wbif_mast1_wrnrdr depending on csr[1] bit to indicate either read or write cycle for wishbone master interface0 or wishbone master interface1. DMA presents dma_wbif_mast0_addr, dma_wbif_mast1_addr, dma_wbif_mast0_odata or dma_wbif_mast1_odata in case of write cycle. DMA may insert wait cycles dma_wbif_mast0_wait or dma_wbif_mast1_wait telling the master that DMA is not ready to accept or provide the data.

In wishbone master interface first Pass through select signal is checked to see for Pass through logic else normal DMA transfer will go. **In case of not a Pass through logic** Wishbone master signals like wishbone address(o_mast0_adr or o_mast1_adr), write enable(o_mast0_we or o_mast1_we), select(o_mast0_byte_sel or o_mast1_byte_sel), strobe(o_mast0_strb or o_mast1_strb) && cycle start (o_mast0_cyc or o_mast1_cyc) are generated using the signals coming from DMA engine.

Master and Slave interface protocol:

Wishbone master 0 or 1 presents o_mast0_adr or o_mast1_adr depending on which master interface is enabled to Slave. Master asserts o_mast0_we or o_mast1_we to indicate a Read/Write cycle. Indicates a write cycle when asserted HIGH. Master presents bank select (o_mast0_byte_sel or o_mast1_byte_sel) to indicate the valid data on the bus, o_mast0_cyc or o_mast1_cyc to indicate the start of cycle, o_mast0_strb or o_mast1_strb strobe when master is ready to transfer data to Slave. This remain asserted until slave asserts o_slv0_ack or o_slv0_err or o_slv0_retry or o_slv1_ack or o_slv1_err or o_slv1_retry. Slave decodes inputs and prepares to latch data on i_slv0_data or

i_slv1_data if it is a write cycle, presents valid data on o_mast0_data or o_mast1_data of Master in case of read cycle.

Slave asserts o_slv0_ack or o_slv1_ack in response to o_mast0_strb or o_mast1_strb from master to indicate valid data. Slave may insert wait cycles before asserting ack. Master monitors o_slv0_ack or o_slv1_ack and prepares to latch data for Read cycle. Master negates o_mast0_strb or o_mast1_strb and o_mast0_cyc or o_mast1_cyc to indicate the end of cycle.

Now DMA may tell master to insert wait states as DMA is not ready to accept data.

Master error(i_mast0_err or I_mast1_err) in wishbone master is generated internally by taking the o_slv0_err or o_slv1_err signal coming from slave. If any error comes during transfer that is reflected back to DMA through dma_wbif_mast0_err or

dma_wbif_mast1_err to tell DMA that an error has occurred during transfer. Master asserts dma_wbif_mast0_drdy or dma_wbif_mast1_drdy is generated internally from i_mast0_ack or I_mast1_ack to tell DMA that data is available during a read or that the master can accept the next data during a write.

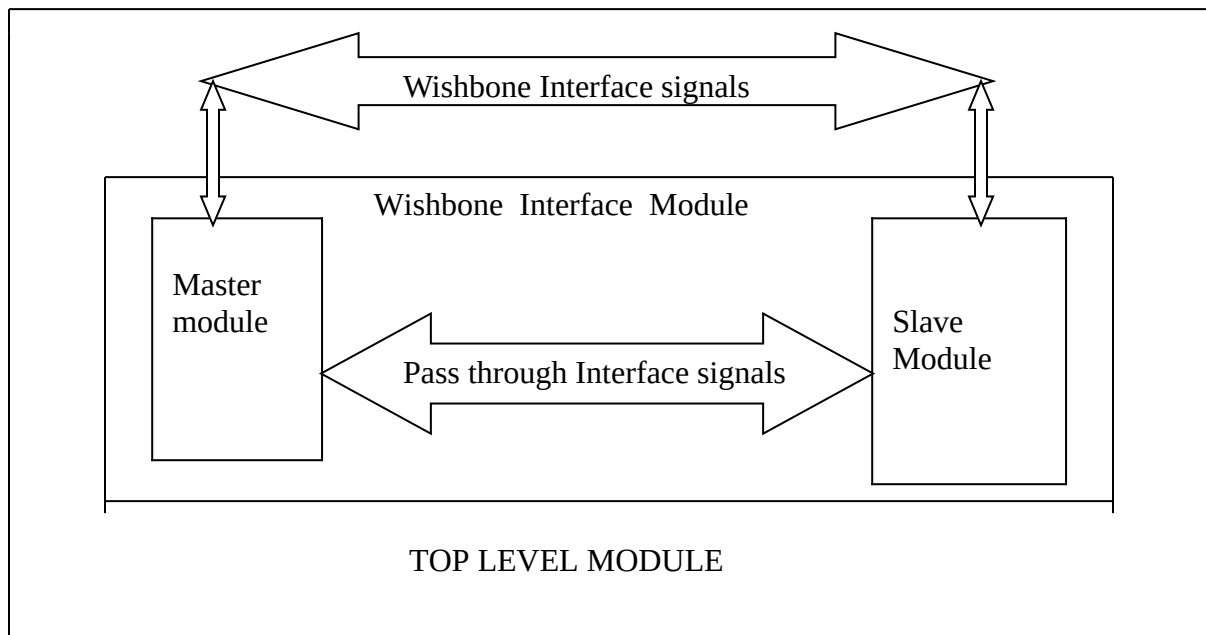
Slave to Regfile interface:

In case of Read/Write signal coming from Master Slave depending on the Pass through signal not selected selects the Register file interface signals for data transfer. Slave puts the address (dma_regfile_addr) and data (dma_wbif_datain in case of write cycle) on the data and address bus indicating to Register file that whether Read/Write cycle is there.

When dma_regfile_wrnrd is HIGH indicates a Write cycle.

Register file interface after reading the data puts the data on to the dma_regfile_dataout bus. After the Read/Write operation Regfile interface sends an dma_regfile_ack acknowledge signal to Slave indicating that the operation is over.

Interface signals in Wishbone:



Wishbone Master and Slave signals communicate with each other through two interfaces

declared. These are Wishbone interface which bundles all the signals of Wishbone Master and Slave signals is decalred in wishbone top module. Another interface is Pass through interface which bundles all the signals of Pass through signals of Master and Slave module decalred in Wishbone interface module.

Appendix:

Signal Name	Width	Description
o_mast0_adr	32	Address output from Master 0
o_mast1_adr	32	Address output from Master 1
o_mast0_data	32	Data output from Master 0
o_mast1_data	32	Data output from Master 1
o_mast0_byte_sel	4	Output from Master 0.Indicates which bytes are valid on the data bus.
o_mast1_byte_sel	4	Output from Master 1.Indicates which bytes are valid on the data bus.
o_mast0_we	1	Output from Master 0. Indicates a write cycle when asserted HIGH
o_mast1_we	1	Output from Master 1. Indicates a write cycle when asserted HIGH
o_mast0_cyc	1	Output from Master 0. Indicates a vlid transfer cycle.
o_mast1_cyc	1	Output from Master 1. Indicates a vlid transfer cycle
o_mast0_strb	1	Output from Master 0. Indicates a valid transfer
o_mast1_strb	1	Output from Master 1. Indicates a valid transfer
i_mast0_data	32	Input data to Master 0
i_mast1_data	32	Input data to Master 1
i_mast0_ack	1	Input for Master 0. Acknowledgement output. Indicates a normal cycle termination
i_mast1_ack	1	Input for Master 0. Acknowledgement output.Indicates a normal cycle termination
i_mast0_err	1	Input for Master 0. Error acknowledgement. Indicates an abnormal cycle termination
i_mast1_err	1	Input for Master 0. Error acknowledgement. Indicates an abnormal cycle termination
i_mast0_retry	1	Input fro Master 0. Retry output. Indicates that the interface is not ready and the master should retry this operation
i_mast1_retry	1	Input fro Master 0. Retry output. Indicates that the interface is not ready and

		the master should retry this operation
o_slv0_data	32	Output data from Slave 0
o_slv0_ack	1	Output from Slave 0. Acknowledgement output. Indicates a normal cycle termination
o_slv1_ack	1	Output from Slave 1. Acknowledgement output. Indicates a normal cycle termination
o_slv0_err	1	Output from Slave 0. Error output. Indicates an abnormal cycle termination
o_slv1_err	1	Output from Slave 1. Error output. Indicates an abnormal cycle termination
o_slv0_retry	1	Output from SLAVE 0. retry output.. Indicates that the interface is not ready and slave 0 should retry this operation
o_slv1_retry	1	Output from SLAVE 1. retry output.. Indicates that the interface is not ready and slave 0 should retry this operation
i_slv0_data	32	Input data to Slave 0
i_slv1_data	32	Input data to Slave 1
i_slv0_addr	32	Address input to slave 0
i_slv1_addr	32	Address input to slave 1
i_slv0_byte_sel	4	Input for slave 0. Indicates which bytes are valid on the bus.
i_slv1_byte_sel	4	Input for slave 1. Indicates which bytes are valid on the bus.
i_slv0_we	1	Input for slave 0. Indicates a write cycle when asserted HIGH
i_slv1_we	1	Input for slave 1. Indicates a write cycle when asserted HIGH
i_slv0_cyc	1	Input for slave 0. Indicates a valid transfer cycle
i_slv1_cyc	1	Input for slave 1. Indicates a valid transfer cycle.
i_slv0_strb	1	Input for slave 0. Indicates a valid transfer.
i_slv1_strb	1	Input for slave 1. Indicates a valid transfer.

Table:1 showing the wishbone Master and Slave Interface signals

Signal names	Width	Description
pt0_sel_in	1	Input pass through signal to master 0. . When HIGH pass through mode is selected
Pt1_sel_in	1	Input pass through signal to master 1. . When HIGH pass through mode is selected
mast0_pt_data_in	71	Pass through data input signal to master 0.
mast1_pt_data_in	71	Pass through data input signal to master 1.
mast0_pt_data_out	35	Pass through data output signal for master 0.
mast1_pt_data_out	35	Pass through data output signal for master 1.
pt0_sel_out	1	Output pass through select signal from Slave 0.
pt1_sel_out	1	Output pass through select signal from Slave 1.
slv0_pt_data_in	35	Pass through input data signal to Slave 0
slv1_pt_data_in	35	Pass through input data signal to Slave 1
slv0_pt_data_out	71	Pass through data output signal to Slave 0
slv1_pt_data_out	71	Pass through data output signal to Slave 1

Table2: Showing the Pass through interface signals For Master and Slave

Signal names	Width	Description
dma_wbif_mast0_start	1	Output from DMA to Master 0 to start a master cycle
dma_wbif_mast0_wait	1	Output from DMA to master 0 to wait for a few cycles as the DMA is not ready to accept/receive the data
dma_wbif_mast0_wrnd	1	Output from DMA to master 0. Indicates a Write cycle when asserted HIGH.
dma_wbif_mast0_drdy	1	Output from master 0 to DMA indicating that either data is available during a read or that the master can accept the next data during a write
dma_wbif_mast0_err	1	Output from Master 0 to DMA indicating that an error has occurred during the transfer
dma_wbif_mast0_addr	32	Address output from DMA to Master 0
dma_wbif_mast0_odata	32	Data output
Dma_wbif_mast0_idata	32	Data input
dma_wbif_mast1_start	1	Output from DMA to Master 1 to start a master cycle
dma_wbif_mast1_wait	1	Output from DMA to master 1 to wait for a few cycles as the DMA is not ready to accept/receive the data
dma_wbif_mast1_wrnd	1	Output from DMA to master 1. Indicates a Write cycle when asserted HIGH.
dma_wbif_mast1_drdy	1	Output from master 1 to DMA indicating that either data is available during a read or that the master can accept the next data during a write
dma_wbif_mast1_err	1	Output from Master 1 to DMA indicating that an error has occurred during the transfer
dma_wbif_mast1_addr	32	Address output from DMA to Master 1
dma_wbif_mast1_odata	32	Data output
dma_wbif_mast1_idata	32	Data input

Table 3: Showing the interface signals of DMA to Wishbone interface Master

Signal names	Width	Description
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i_rd		Read signal from Slave to Regfile
i_wr_datain	32	Data input to regfile from Slave
i_wr_addr	32	Address input to regfile from Slave
i_wr	1	Write signal from Slave to Regfile
regfile_wbif_data	32	Data output from regfile to Slave

Table4: Showing the interface signal os Slave with the Register File

Signal names	Width	Description
i_clk	1	Input clock
i_reset	1	Input Reset

Table 5: Showing the additional signals